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# Supplementary Materials for

## The Effect of Treatment Expectation on Drug Efficacy: Imaging the Analgesic Benefit of the Opioid Remifentanil

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**The PDF file includes:**

# Supplementary Materials

## Supplementary Methods

### Exclusion of four volunteers

Twenty-six volunteers were recruited and attended the introductory visit. Of these volunteers, two experienced opioid-induced nausea and were excluded from the rest of the study. One volunteer completed the introductory session but was unable to attend the main experimental session because she contracted an upper respiratory tract infection. The remaining 23 volunteers participated in the main experimental session performed with fMRI. Of these participants, twenty-two completed the study. One participant withdrew from the main experimental session because of claustrophobia.

### Introductory Session

This behavioural session was performed at least 24h before the main experimental session. The session was designed to ensure that the participants tolerated all experimental procedures, including the remifentanil infusion and to introduce the experimental paradigm used during the main experimental session performed with fMRI. It also included a conditioning-like procedure to reinforce positive and negative treatment expectancies (see below). The conduct of the introductory session was as follows:

*Instructions.* The same experimenter (U.B.) welcomed each participant and obtained written informed consent for the study. The participant was told that the aim of the study was to 'find out why people respond differently to opioids'. The participants were informed that remifentanil is a widely used opioid that relieves provoked pain when infused intravenously, but can worsen provoked pain when the infusion ceases (S1). Participants had given consent to not always being aware of exact experimental procedures including the application of remifentanil during the course of the experiment. An anaesthetist (V.W. or R.M) ensured that the participants were healthy (S2), and confirmed that the participant had no contraindications to remifentanil or to magnetic resonance imaging. The participants then completed the following questionnaires during the Introductory Visit: Trait and State anxiety (S3), Center for Epidemiologic Studies Depression Scale (S4), Fear of Pain Questionnaire-III (S5), PVAQ (S6), Lot-R (S7), BISBAS (S8), CATS (S9) and Levenson's Internal, Powerful Others, and Chance locus of control scales (S10).

*Pain thresholding.* The threshold to pain evoked by contact-heat stimuli (30 x 30mm ATS thermode, Pathway System, Medoc, Israel) applied to the lateral aspect of the right mid-calf was determined using the method of limits in each participant. We only included participants with heat pain thresholds that lay within one standard deviation of the published normative data provided by the manufacturer of the thermode, which are in part published in (S11, S12) and (S13). The participant was familiarized with the pain sensations and the use of the visual analogue scale (VAS) for reporting pain intensity (VAS; 100 parts; endpoints labelled with "no pain" and "unbearable pain"). For each participant, the stimulus temperature that was rated at ‘70’ on the VAS was determined and selected for use at the start of the experimental paradigm.

*Experimental paradigm.* Each participant received a total of forty painful thermal stimuli divided into four runs – two baseline runs, one remifentanil run and one remifentanil-stopped run. The first two runs (baseline 1 and 2) were completed with a saline infusion only. Then, the remifentanil infusion (see Drug Administration) was started. The participant was asked to wait for ten minutes “to allow the drug to fully exert its action”, before the third run (“remifentanil run”) was performed. Upon completion of the third run, the infusion of remifentanil was terminated. The participant was asked to wait for another ten minutes “to allow the drug to clear from the system”, before the fourth run (“remifentanil stopped”) was commenced. The waiting times were not related to any pharmacokinetic considerations but were included so that a plausibility account of opioid effects can be given to the participants even as their expectations were manipulated experimentally. These timings were repeated during the fMRI experiment for consistency. The experimental design of the introductory session is illustrated in Fig. S1.

*Conditioning-like procedure.* During both baseline runs, the temperature corresponding to a VAS-score of 70 was used. To reinforce the participant’s expectation of analgesia during remifentanil infusion, the stimulus temperature was decreased by 1C° during the “remifentanil run”. Likewise, to enhance the expectation of hyperalgesia after cessation of the infusion, the stimulus temperature was increased by 1C° during the “remifentanil-stopped” run. The participant was unaware of these temperature changes. The participant was informed about the experimental condition by the words “baseline1”, “baseline2”, “remifentanil infusion”, or “remifentanil stopped” which were displayed in the right upper quadrant of the computer screen throughout the according run.

The participants were told that the same experimental paradigm would be used in the fMRI session. In fact, the experimental procedures performed during fMRI differed in two respects (see Figure 8A). First, the same temperature corresponding to a pain intensity of 70/100 was used throughout all four runs. Second, the remifentanil infusion commenced after the first baseline run and continued throughout the experiment without the participant knowing. This allowed us to examine the analgesic efficacy of a constant dose of remifentanil under three different conditions, namely no expectation, positive expectation and negative expectation (see also main manuscript).

**Study Drug**

The opioid remifentanil used in this study is a potent, synthetic μ-opioid agonist with a rapid onset of action, a context sensitive half life of 3-4 min (S14) and an elimination half life of ~ 10 minutes (S15). These properties make it ideal for healthy volunteer experimental studies where rapid onset and offset of opioid action is required. Furthermore, other groups studying the effects of opioid modulation of pain in healthy volunteers have also used remifentanil (S16). Its rapid onset of activity with a short half-life, has distinct advantages as a tool in experimental pain studies. In terms of volunteer safety, it is possible to achieve a relatively strong opioid effect with remifentanil, but if adverse effects develop, the infusion can be stopped with the confidence that the drug will wear off within minutes. The common side effects are respiratory depression, nausea and vomiting. Because of the unique pharmacokinetic properties of remifentanil it needs to be given as an intravenous infusion and it is possible to achieve target plasma and effect site (brain) concentrations in a predictable and consistent manner. We used a target controlled infusion pump to deliver the remifentanil to achieve the desired target effect site concentration using the method outlined in drug administration section.

Please note, the dose of remifentanil used in our study is a clinically relevant, albeit in the lower range. It falls within the range of infusion rates (0.027 – 0.207 μg/kg/min) required for the relief of acute severe clinical pain (S17, S18) and may be assumed to have comparable analgesic effects to non-steroidal anti-inflammatory drugs (NSAIDs) that are widely used to treat low to moderate pain.

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**Drug administration**

All participants fasted for 6 hours before each of the two visits and were monitored for an hour after termination of the infusion. A target-controlled infusion of remifentanil (at a solution concentration of 10μg/ml) was delivered via an indwelling intravenous cannula inserted in the left forearm. The estimated effect site concentration of remifentanil used during the fMRI sessions was 0.8 ng/ml. A slightly higher concentration of remifentanil (1.0 ng/ml) was used in the shorter introductory session for the detection of adverse effects that would contraindicate fMRI. The desired effect-site concentrations were achieved by combining a bolus with a maintenance dose delivered via a pump (Graseby 3500 TCI incorporating

Diprifusor; SIMS Graseby) that was pre-programmed according to a pharmacokinetic model of remifentanil that included the participants weight, height and gender (S19, S20). The choice of these doses was based on previous remifentanil studies conducted in our laboratory (S21, S22). Preliminary data indicate that this regimen produced consistent reports of analgesia across subjects, with minimal psychotropic effects that can be easily dismissed and attributed to the noisy scanner environment (also see control experiment 2 and Fig S5).

**Physiological measurements**

An anaesthetist was present throughout both experimental visits and monitored the participant's physiology. Oxygen was administered at a flow rate of 1 litre per minute via nasal cannulae with prongs which allowed simultaneous oxygen delivery and sampling of expired airway gases (Salter Labs). Heart rate, peripheral oxygen saturation ($SpO_2$), respiratory rate, end-tidal $CO_2$ and partial pressures ($P_{ET}CO_2$) were measured continuously (9500 Multigas Monitor, Wardray Premise). Manual and electronic records were maintained in order to fulfil subjects' monitoring requirements and allow comparison of $P_{ET}CO_2$ levels across runs (see supplemental Results). Non-invasive arterial blood pressure was recorded before and during the remifentanil administration. To prevent the effects of opioid induced hypoventilation and hypercarbia, particularly on BOLD responses during the fMRI experiment, the participant was trained and constantly reminded (by visual cues on the screen) to breathe regularly throughout the experiment regardless of whether saline or remifentanil was administered.

**Post-hoc confidence questionnaire**

To assess potential unblinding, the participants' confidence in the actual experimental condition was evaluated by a post-hoc questionnaire, which the participants completed after the fMRI scan before they were debriefed regarding the actual experimental procedures. This questionnaire contained the following questions:

1) How satisfied were you with the analgesic effect of remifentanil?
2) Would you recommend remifentanil to a friend e.g. when undergoing a medical procedure?
3) How confident were you that you had received remifentanil during the remifentanil session?
4) How confident were you that you did not receive remifentanil during the baseline sessions?
5) How confident were you that you did not receive remifentanil during the remi-stopped session?

Each question could be answered on a visual analogue scale where 0 denoted "not at all" and 100 "absolutely".

**FMRI data acquisition**

Functional magnetic resonance imaging (fMRI) data were acquired on a 3 Tesla system (Varian, Siemens). 42 transverse slices (slice thickness 3mm with 3 x 3 mm in-plane resolution, repetition time was 3 sec, echo time 30 msec, field of view 224 x 224 mm, matrix size 64 x 64 pixels) were acquired using T2*-weighted echo-planar imaging (EPI). The influence of in-plane susceptibility gradients was reduced by tilting the imaging slice by 30 degree from axial to coronal orientation. The first four volumes were discarded to permit equilibration of the BOLD signal. We also acquired high-resolution (1x1x1mm voxel size) T1-weighted images for each subject using a MP-RAGE sequence.

**Data analysis – fMRI**

fMRI data processing and statistical analyses were carried out using statistical parametric mapping (SPM5, Wellcome Trust Centre for Neuroimaging, London, UK). Data processing consisted of slice timing correction (for differences in slice acquisition time), realignment (motion correction), unwarping, spatial normalization to a standard EPI template and smoothing with an 8mm (FWHM) isotropic Gaussian kernel. Data were also subjected to high-pass filtering (cutoff period: 128s) and correction for temporal autocorrelations (based on a first-order autoregressive model).

Data analysis was performed using a general linear model approach. For each subject the first level design matrix included 4 experimental runs, each of which had 6 regressors, which modelled the predicted BOLD responses to the different events that comprised one pain trial (Fig. 8C). They were (i) the presentation of the reaction time cue, (ii) presentation of the reaction time target, (iii) the pain anticipation phase, (iv) painful thermal stimulation, (v) the pain rating procedure and (vi) the visual control stimulation. All regressors were obtained by convolving event duration with the canonical hemodynamic response function implemented in SPM.

After model estimation, the ensuing first-level contrast images from each subject were used for second-level analyses, applying SPMs “Full factorial” model. For the 2$^{nd}$ level analyses, the model comprised four regressors reflecting the BOLD response during painful thermal stimulus, one for each run (baseline, no expectancy, positive expectancy, negative expectancy). Inferences were made by entering the appropriate contrast into the ANOVA.

We corrected for possible non-sphericity of the error term. The main aim of this study concerned neural activity during painful thermal stimulation. As such, data from the other regressors are not presented.

We investigated the following effects sequentially. First, we confirmed that painful stimulation in our experiment activated the putative brain regions responsible for pain processing and perception. For this analysis, we generated the statistical map of brain activation during pain for the first run (saline application) only. Next, we tested for the intrinsic analgesic effect of remifentanil analgesia, which we define as the effect of drug when no drug was expected. As such, we contrasted the pain activations of run 1 (saline) with run 2 (covert application of remifentanil, no expectancy).

We then isolated the regions of the brain where activity correlated with perceived pain intensity, regardless of the context under which pain was experienced. The contrast weights were z-transformed mean ratings of pain intensity from all four experimental runs. Activity changes observed in core regions of pain processing brain areas would provide strong objective evidence beyond subjective ratings that expectancies regarding the efficacy of an analgesic can critically affect its analgesic potency.

To confirm the above result, i.e. that the reported changes in analgesia during the different expectancy conditions are associated with activation changes in pain-related brain areas, we took an additional approach. Therefore, we identified all regions that showed significant BOLD responses to painful stimulation during the baseline (saline infusion) run, see Fig. 3 and Table S1. We then extracted parameter estimates from each of these regions (6 mm sphere around peak voxel) for the three experimental runs where remifentanil was given under different expectancies (no expectation, positive expectation and negative expectation) from each subject. We then calculated an average parameter estimate across all regions per subject that was then entered into a repeated measures ANOVA and subsequent post-hoc T-Tests to compare brain responses in brain-related brain regions between the three remifentanil conditions. As we used one part of the data (first run) to identify the voxels activated by pain, which where then used to perform the subsequent analysis of analgesia during the three different expectancy conditions (run 2-4) this approach does not suffer from circularity (S*23*).

Next we turned to the clinically most relevant question of how positive or negative expectations of treatment outcome modulate pain sensitive brain areas and the brain circuitry subserving opioid analgesia. Therefore, we compared brain responses to identical pain stimuli during the negative expectancy run and the positive expectancy run.

First, we then examined which brain regions showed increased activation during positive compared to negative expectancies (run 4 > run 3). This analysis served to confirm increased activations in brain regions involved in the perception of pain intensity, as well to identify activations in regions that can facilitate pain and decrease the analgesic potential of opioids.

The reverse analysis [run 3 > run 4) was used to identify brain areas showing increased activation when remifentanil was received with positive expectancy. These areas represent the modulatory network that mediates the increased analgesic potency of opioids during positive expectancy.

Finally, we determined where brain activations might predict the influence of negative and positive expectancies on opioid analgesia in individuals. To this end we performed correlation analyses of the change in activation between positive and negative expectancy by using the standard regression analysis implemented in SPM05, using the individual behavioural difference between the positive and negative expectancy condition as a covariate. Correlation coefficients are reported for a 6 mm sphere around the peak voxels of these analyses.

We used a threshold of $p \leq 0.001$ uncorrected. We also report corrected p-values as obtained from small volume correction in a-priori regions of interest at a level of $p \leq 0.05$. These include brain areas known to be involved in pain processing (brainstem, thalamus, basal ganglia, insula, somatosensory and midcingulate cortex (S*24*)) and its modulation (prefrontal cortex, rostral anterior cingulate cortex (rACC), amygdala and hippocampus (S*25, S26*)). Correction was based on spheres centred around the peak coordinates (ignoring laterality) obtained from previous studies (S27-S29). Activation in cortical areas were corrected using spheres of 20mm, subcortical areas were corrected using spheres of 10mm.

## Supplementary Results

### Behavioural Results

*Correlation analyses.* The subsequent correlation analyses comprised the following calculations: The intrinsic analgesic potential of remifentanil (no expectancy) was assessed by subtracting the pain intensity ratings of the no-expectancy run from those of the baseline run. The effect of positive treatment expectancy was calculated as the difference in pain intensity ratings between no expectancy and the positive expectancy run. The effect of negative treatment expectancy was calculated as the difference in pain intensity ratings between the positive expectancy and the negative expectancy run.

We first investigated whether the analgesic effect during the covert administration of remifentanil, representing the intrinsic drug effect, correlated with the additional analgesic benefit from positive expectancy. This was not the case ($r=-0.15$, $p = 0.25$). However, the pain relieving/aggravating effects of positive and negative expectation showed a significant correlation ($r = 0.58$, $p < 0.001$). In other words, participants with a high analgesic benefit from positive expectancy also experienced a stronger increase in pain ratings in the 4$^{th}$ run, when participants expected the pain to become worse. However, note that our design does not allow for assessing these measures independently.

*Scores from the personality questionnaires:* The intrinsic effect of remifentanil analgesia showed a trend level positive association with the bas reward score ($r=0.3$, $p=0.1$) and the bas total ($r=0.3$, $p=0.1$).

No correlation was found between the effect of positive treatment expectancy and any of the personality scores.

The pain unpleasantness ratings during the baseline run (in which no opioid was given) highly correlated with the individual effect of negative treatment expectancy ($r = 0.52$, $p = 0.006$). Intriguingly, high scores in optimism (as obtained by the LOT-R) were associated with lower expectancy ratings of pain increase in the 4th fMRI run ($r = -0.4$, $p = 0.03$). However, these optimistic expectancies did not prevent the optimists from the experience of pain increase in the 4th run, as optimism was not negatively correlated with the actual increase in pain intensity from the 3rd to the 4th run ($r = -0.13$, $p = 0.28$). Higher depression values (as obtained by the CESD score) were associated with higher expectancy ratings for the pain to increase in the 4th run ($r = 0.36$, $p = 0.05$, trend level). But again, CESD levels did not significantly effect the actual development of pain increase in the 4th run ($r = 0.25$, $p = 0.13$).

**fMRI data**

*Pain related brain areas.* Activity in pain related brain areas (identified during baseline) during different expectancies of opioid analgesia. The ANOVA analysis of the parameter estimates shows that the expectancy modulation of opioid analgesia is present across the whole network of brain areas activated by pain [$F(1.4,26.0) = 20.8$, $p<0.001$, Greenhouse-Geisser corrected for non-sphericity]. Post-hoc tests showed that positive expectancy significantly enhanced analgesia as pain-related activity decreased from the no expectancy to the positive expectancy condition [$t(19) = -5.6$, $p < 0.001$]. Negative expectancy, when the subjects had been led to believe that the drug was stopped, resulted in an increase in pain related activity [$t(19) = 7.1$, $p < 0.001$](Fig S2).

*Brain areas mediating the effects of positive and negative expectancy, inspection of the parameter estimates.* Comparing brain responses to identical pain stimuli under conditions of negative expectancy and under conditions of positive expectancy showed that the attenuated analgesic effect (i.e. increase in pain intensity) during negative expectancy is associated with an increase in brain activity in the hippocampus bordering the amygdala, medial prefrontal cortex and the cerebellum (Table 2, Fig. 6). In contrast, increased analgesia under positive expectancy was associated with activity in the dorsolateral prefrontal cortex, ACC (including rostral and peri/sub-genual aspects), the striatum, including caudate nucleus and putamen, and the frontal operculum (Table 2, Fig 7).

To further unravel which condition is actually driving these effects, we extracted the parameter estimates from these areas identified to be associated with positive and negative activity and known from published literature to be relevant for driving placebo analgesia and nocebo hyperalgesia (subgenual ACC (sgACC) and hippocampus, respectively, 6 mm sphere around the peak voxel from the group statistics) from each subject for all three remifentanil conditions. These were entered into repeated measures ANOVA and subsequent post-hoc T-Tests performed. This analysis enables inferences to be made regarding increases and decreases in BOLD activity during each of the three remifentanil conditions.

Inspection of these parameter estimates reveals increased activity in the sgACC when analgesia is increased during positive expectancy and a deactivation of this region when analgesia is impaired during negative expectancy [ANOVA $F(2,42) = 7.3$, $p<0.01$, post hoc T-Tests $t(21) = -2.3$, $p<0.05$ for run 2 vs. run 3, $t(21) = 3.9$, $p < 0.01$ for run 3 vs. 4](Fig. S3, right). This response pattern suggests that both positive and negative expectancy use a key component of the descending pain modulatory control system, but in opposite ways.

Inspection of the parameter estimates shows no response in the hippocampus when remifentanil is applied in the no-expectation or the positive expectation condition, but a strong increase in activity occurs when analgesia is impaired during negative expectancy

[ANOVA $F(1.53,32.2) = 3.9$, $p<0.05$, Greenhouse-Geisser corrected for non-sphericity, post hoc T-Tests $t(21) = -0.3$ for run 2 vs. run 3, p = n.s., $t(21) = -3.3$, $p < 0.01$ for run 3 vs. run 4] (Fig. S3, left). This response pattern indicates the selective contribution of the hippocampus to the mechanisms of negative expectancy (nocebo).

**Physiological Data**

For technical reasons physiological data is only available from 18 out of the 22 participants.

*Pulse Rate.* Consistent with the known cardiovascular effects of remifentanil, pulse rate slightly decreased after the beginning of the remifentanil infusion from 60/min during baseline to 57/min during the covert remifentanil run ($t(17) = 3.3$, $p = 0.005$). Manipulation of treatment expectancies did not affect the pulse rate (pulse rates run 3 = 56/min, run 4 = 56/min, no significant differences).

*End-Tidal $CO_2$ and $SpO_2$.* The repeated measures ANOVA revealed a significant effect of experimental condition on $CO_2$ ($F(3,48) = 8.5$, $p < 0.001$). Subsequent post-hoc tests showed that the end-tidal $CO_2$ increased slightly from the baseline run to the first remifentanil run (40.7 to 43.7, $t(17) = 5.0$, $p <0.001$). Importantly, no change in end-tidal CO2 during the three remifentanil runs was observed (end-tidal CO2 run 3 = 44.0, run 4 = 43.2, no significant differences). In all subjects the peripheral oxygen saturation ($SpO_2$) levels did not drop below 98 throughout the experiment.

**Post-hoc confidence questionnaire**

The mean ratings ± sem were: question 1: 76 ± 7, question 2: 68 ± 7, question 3: 82 ± 7, question 4: 67 ± 8, question 5: 74 ± 6.

**Control experiments**

Two control experiments were performed with procedures and timing identical to the fMRI part of our study to assure that context dependent changes in pain ratings are not attributable to sensitization or habituation phenomena, or time dependent changes of opioid analgesia.

*Control experiment 1: Habituation and sensitization over time*

Ten additional participants (6 male, 4 female, mean age 26 years) were enrolled in this behavioural pilot experiment. 40 identical thermal stimuli tailored to evoke a pain rating of 70/100 on a VAS were applied in four runs, with stimulus timings that were identical to the fMRI part of the study. Importantly, the stimulation site was slightly moved along the right mid-calf between runs, as in the introductory and the fMRI session of the study. As illustrated in Fig. S4, no habituation or sensitization phenomena occurred with our stimulation paradigm.

*Control experiment 2: Natural time course of remifentanil analgesia*

This control experiment was performed to test that the remifentanil regimen used in the fMRI part of our study results in stable analgesia over the entire course of the experiment. This was tested in 9 further participants (5 male, 4 female, mean age 25 years). The time course and all experimental procedures (including painful thermal stimulation, remifentanil dose, physiological monitoring, etc.) were identical to the main experimental session of our study. However, no contextual manipulation of the drug efficacy was performed. As illustrated in Fig. S5, the remifentanil regimen did not lead to significant change in pain intensity rating between the three remifentanil runs indicating stable analgesia over the course of the experiment. Importantly, this indicates that the effects of treatment expectancies of the analgesic potential of remifentanil in our study can not be ascribed to the effect of time or prolonged infusion of remifentanil.

*Posthoc Analysis of the time course of changes in analgesia during the different expectancy conditions.*

To further substantiate that our results are not biased by habituation, sensitization or other time dependent phenomena we have plotted the time course of pain ratings to each of the painful stimuli applied in the experiment. Fig. S6 illustrates that pain ratings remain unaltered within one experimental run, but step-wise changes of pain ratings occurred between the experimental conditions tightly linked in time with the expectation manipulation. This time pattern strongly supports that the observed changes in analgesia are the result of the expectancy modulation and not of habituation or sensitization phenomena. For these phenomena gradual changes over time would be expected, also occurring within one experimental run.

## Supplementary Figures

### Supplementary Figure 1

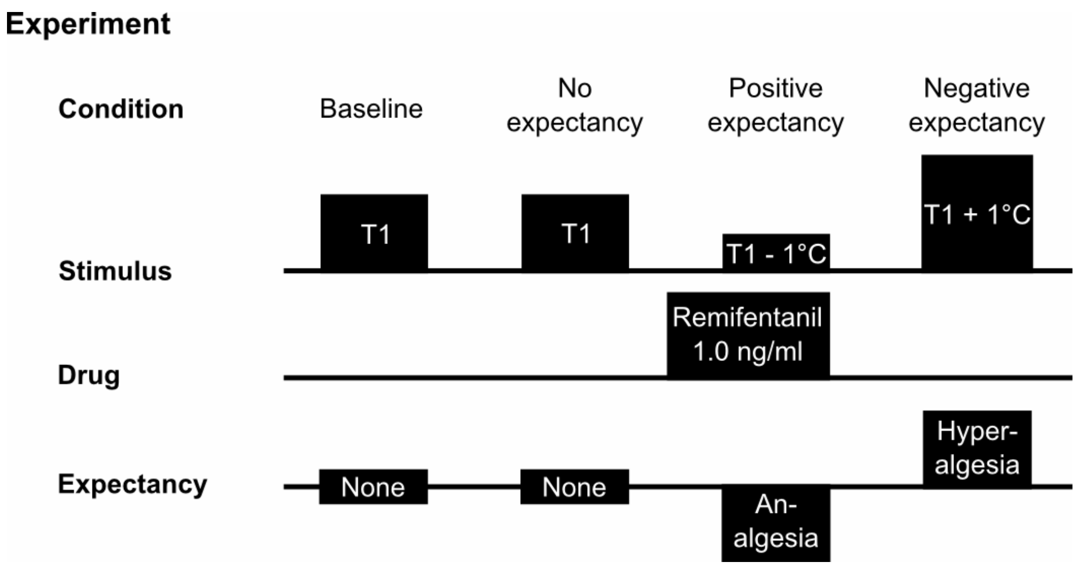


**Fig. S1. Experimental design during the introductory session.** The participant received a total of forty painful thermal stimuli, divided into four runs – two baseline runs, one remifentanil run and one remifentanil stopped run. The first two runs were completed with a saline infusion only. The remifentanil run was performed 10 minutes after commencement of the remifentanil infusion (see Drug Administration) and the remifentanil-stopped run ten minutes after termination of remifentanil infusion. The participant was explicitly informed when remifentanil infusion commenced or ceased. To reinforce the participant's expectation of the analgesia during remifentanil infusion, the stimulus temperature was lowered by 1C° during the remifentanil ($3^{rd}$) run. Likewise, to enhance the expectation of hyperalgesia after the infusion ceased, the stimulus temperature was raised by 1C° during the remifentanil-stopped ($4^{th}$) run. The participant was unaware of these temperature changes. The current experimental condition (baseline1, baseline2, remifentanil infusion, or remifentanil stopped) was displayed continuously in the right upper quadrant of the computer screen. The visual display and the timings of thermal stimulation during runs and trials were identical to those used in the main experimental session (Fig. 1 main manuscript).

**Supplementary Figure 2**

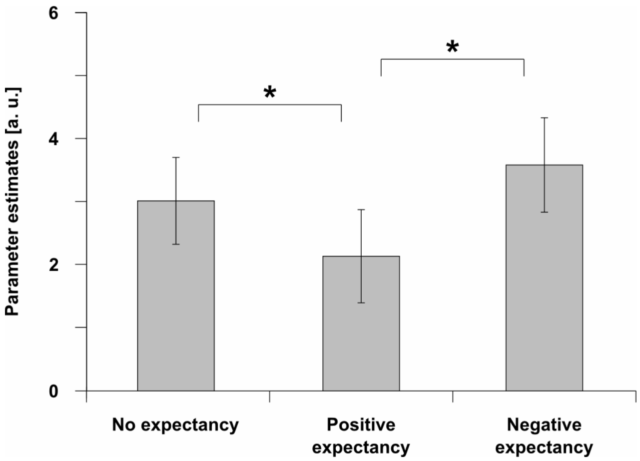


**Fig. S2. The effect of expectancy modulation of opioid analgesia in the core regions of the pain neuromatrix.** Parameter estimates of pain related BOLD-responses (averaged across pain responsive regions as identified during the baseline session, Table S1) during opioid analgesia under the three different expectancy conditions. Error bars indicate the standard errors of the mean (SEM); * = p < 0.05; a.u. = arbitrary units

**Supplementary Figure 3**

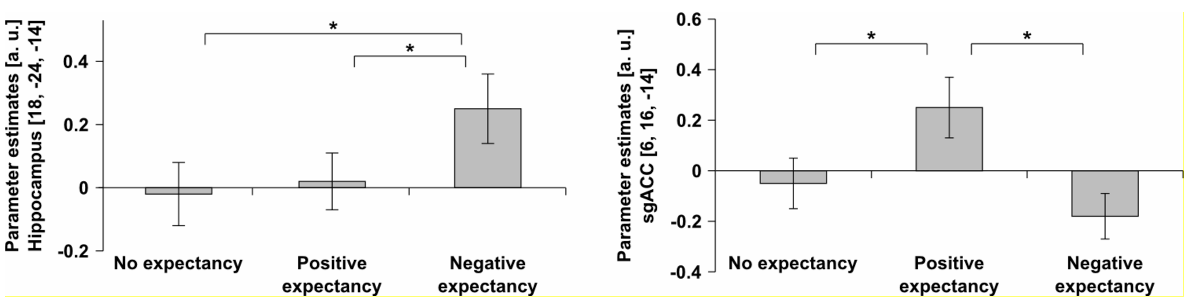


**Fig. S3. Brain areas mediating the effects of positive and negative expectancy. (Left)** Response pattern in the hippocampus during the three remifentanil conditions. Inspection of the parameter estimates shows no response in the hippocampus when remifentanil is applied in the no-expectation or the positive expectation condition, but a strong increase in activity occurs when analgesia is impaired during negative expectancy. **(Right)** Response pattern in the subgenual ACC during the three remifentanil conditions. Inspection of these parameter estimates reveals increased activity in the sgACC when analgesia is increased during positive expectancy and a deactivation of this region when analgesia is impaired during negative expectancy. Error bars indicate the standard errors of the mean (SEM); * = p < 0.05; a.u. = arbitrary units.

**Supplementary Figure 4**

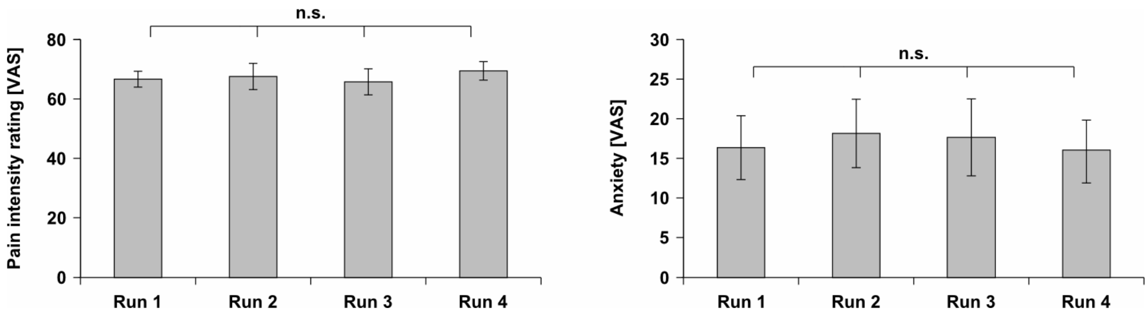


**Fig. S4: Control experiment I—exclusion of habituation or sensitization effects.** Mean pain intensity ratings (left panel) and anxiety ratings (right panel) obtained on the VAS [0-100] for the 4 runs of control experiment 1. Here, 40 identical thermal stimuli were applied with timings that were identical to those of the main experimental session performed with fMRI without any expectancy manipulation with a saline infusion only. Note that the stimulation procedure used in our experiment results in consistent pain intensity ratings over the course of the experiment, when no expectancy manipulation is performed. Error bars indicate the standard errors of the mean (SEM). n.s. = not significant

**Supplementary Figure 5**

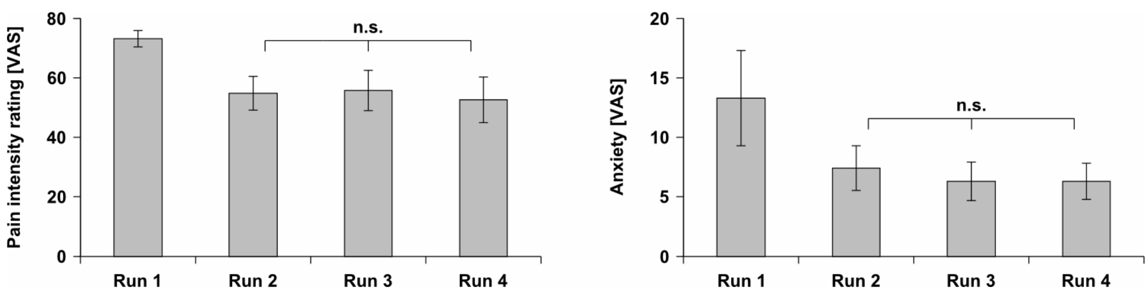


**Fig. S5: Control experiment II—the natural time course of remifentanil analgesia without expectancy manipulation.** Mean pain intensity ratings (left panel) and anxiety ratings (right panel) obtained on the VAS [0-100] for the 4 runs of control experiment 2. This was performed with a constant infusion of remifentanil (effect site concentration 0.8ng/ml) from run 2 to run 4 without any manipulation of treatment expectancy. All other procedures (e.g. thermal stimulation) and timings were identical to those in the main experimental session of our study. Note that this remifentanil regimen led to stable analgesia over the course of the entire experiment, which excludes the confounding effects of prolonged infusion in our experiments. Error bars indicate the standard errors of the mean (SEM), n.s. = not significant

**Supplementary Figure 6**

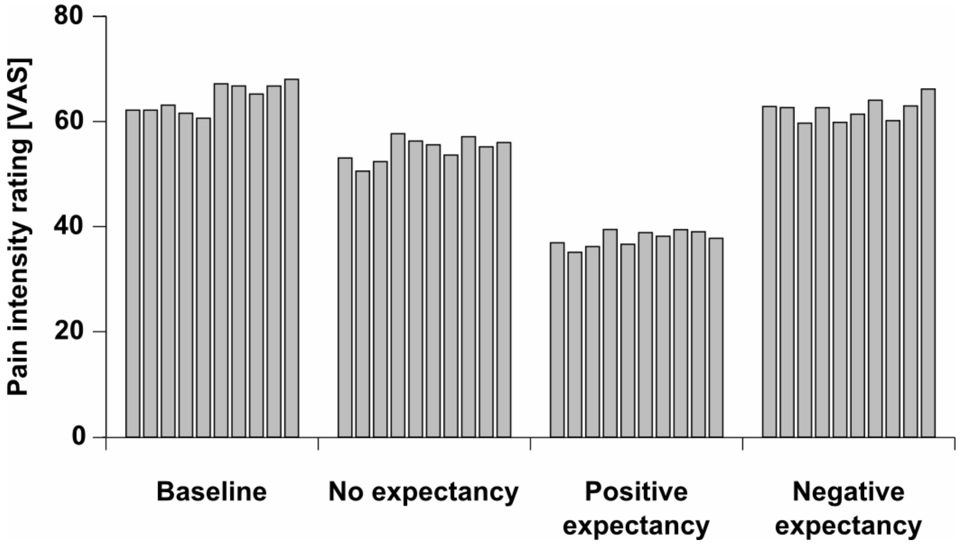


**Fig. S6: Analysis of the time course of changes in analgesia during the fMRI experiment.** Mean pain intensity ratings (across subjects) for each of the 40 pain stimuli applied during the four experimental sessions (10 stimuli each) of the fMRI experiment.

## Supplementary Tables

**Table S1: Effect of painful thermal stimulation.**

| Region | Coordinate of Peak Voxel R | L | Voxel-level (T) R/L |
|---|---|---|---|
| SI | 16, -44, 70 | -12, -44, 70 | 9.0*/14.8* |
| SII | 52, -26, 28 | -60, -22, 24 | 13.1*/14.1* |
| MFG | 38, 42, 21 | -34, 36, 24 | 9.8*/7.9* |
| OFG | 22, 44, -14 | -22, 38, -14 | 6.5*/6.0* |
| ACC | 6, 10, 42 | 6, 14, 36 | 15.9*/17.5* |
| Insula | 36, 6, 8 | -34, 4, 1 | 20.0*/19.4* |
| Striatum | 12, 10, 2 | -12, 8, 0 | 8.2*/7.5* |
| Thalamus | 12, -8, 2 | -16, -12, 8 | 6.3*/7.9* |
| Cerebellum | 36,-68,-2 | -26,-68, -34 | 6.6*/5.7* |
| Brainstem | 10, -22, -16 | -6, -24, -16 | 5.0*/3.9 |

BOLD responses to painful thermal stimulation during the first run of thermal stimulation (saline administration). Coordinates are denoted by x,y, z in mm (MNI-space) and strength of activation is expressed in t-scores (df=63). All p < 0.05 corrected (*), using SVC as indicated in the Supplementary Methods section, or 0.001 uncorrected.

**Table S2: Intrinsic effect of remifentanil on painful thermal stimulation.**

| Region | Coordinate of Peak Voxel R | L | Voxel-level (T) R/L |
|---|---|---|---|
| **Decreased responses** | | | |
| SI | 16, -44, 70 | -8, -34, 72 | 8.0*/5.0* |
| SII | 52, -32, 34 | -44, -30, 30 | 5.0*/3.9 |
| MFG/ DLPFC | | -36, 46, 24 | /4.0* |
| ACC | 6, 22, 28 | -6, 16, 36 | 6.8*/6.1* |
| Insula | 40, 22, 0 | -38, 18, -4 | 6.8*/7.0* |
| Thalamus | | -8, 0, 0 | /3.8* |
| Putamen | 30, 4, 6 | -30, 12, 6 | 6.2*/7.8* |
| Caudate Nucl. | 12, 12, 4 | -14, 14, 4 | 4.4*/4.5* |
| Cerebellum | 20, -38, -30 | -4, -68, -32 | 5.4*/4.9* |
| Brainstem | | -6, -24, -16 | 3.8 |
| **Increased responses** | | | |
| VMPFC | 2, 56, -4 | | 4.9/ |
| Precuneus | | -4, -50, 32 | /5.0 |
| Medial temporal gyrus | 62, -48, 6 | -54, -58, 16 | 4.8/4.2 |

Changes in BOLD responses to painful thermal stimulation during the covert application of remifentanil compared to the baseline run. Coordinates are denoted by x,y, z in mm (MNI-space) and strength of activation is expressed in t-scores (df=63). All p < 0.05 corrected (*), using SVC as indicated in the Supplementary Methods section, or 0.001 uncorrected.

ACC: anterior cingulate cortec, DLPFC = dorsolateral prefrontal cortex, MFG: mid frontal gyrus, OFG: orbitofrontal gyrus, SI: primary somatosensory cortex, SII: secondary somatosensory cortex, VMPFC: ventromedial prefrontal cortex

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